

EE/CprE/SE 491 WEEKLY REPORT 2

9/21 – 9/25 Group: 28

Project title: Biomimetic Prosthesis with Precise Recreation of Hand Movement

Client &/Advisor: Erik Hrysyk, Meng Lu

Team Members/Role:

Reid Post – Powertrain Researcher

Redwan Elshafiey – Sensors and Control Researcher

Logan Ward – EMG Researcher

Adriana Cedillo Sanchez – Anatomy Researcher

Jacob Page – EMG and Materials Researcher

Erik Hrysyk – EMG/powertrain Researcher

Quinn Snyder – PCB/Electronics Lead

Weekly Summary

This week was another week of research. Each member brought ideas to the table for each of the 3 meetings that we attended. The main discussion was over the novelty of the project, and it was mentioned that last year, there was a largely successful senior design project that implemented a mostly functional hand. This could change this year's project into an extension of last year's senior design project sdmay26-41. We are reaching out to the faculty member from that project to see the feasibility of this. This leaves the group at a crossroads, where if we are able to utilize parts or all of their project, then we would be able to slightly redesign it and build upon the lessons that they had learned. If we are not able to get any of their project, we will continue forward with the plan to build our own hand from the ground up, with hopeful stretch goals of providing better user feedback, such as pressure sensors and temperature sensors, as well as some sort of indicator to the user.

Past week accomplishments

- We confirmed more of the scope about what the project will be and have an action plan for when we can get in touch with the mentor for the sdmay26-41 group.

Pending issues

- Confirm project scope

- Buy EMG sensors
- Determine microcontroller
- Figure out underactuation strategies
- We have a large amount of I/O ports
- We do not have novelty in this project, and we may need to redefine our project
- We do not have an EMG band, and it may wipe out our budget
 - o If we are buying an EMG band, we will have to reduce budget for everything else
 - o If we can borrow an EMG, are we then having to worry about any other legal issues, or will we only have limited access to this band
- We do not have a project timeline, largely because we do not have the actual project scope defined
- We do not have our group properly split into sub-groups so that we can make progress quicker
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o **Individual contributions**

<u>NAME</u>	<u>Individual Contributions</u> <i>(Quick list of contributions. This should be short.)</i>	<u>Hours this week</u>	<u>HOURS cumulative</u>
Logan Ward	Brainstorming, group organization, started on EMG research, found potential sEMG's to purchase	6	12
Reid Post	Created document for potential demonstration. General groundwork research for how underactuation works with a prosthetic hand. Provided input on different directions to go depending upon the previous year's work	6	12
Redwan Elshafiey	Brainstorming, group organization, start on sensor research, researching components	6	12
Erik Hrysyk	Brainstorming, Group planning, role selection, start on EMG research	6	12
Jacob Page	EMG Research, Movement and control research, scope identification	6	12

Adriana Cedillo	Brainstorming, researching anatomy and hand movement	6	12
Quinn Snyder	Brainstorming, group organization, start on materials research	6	12

- **Comments and extended discussion** *(Optional)*
- **Open Action Items**
 - Find open-sourced Github EMG model for gesture recognition
 - Buy/Find EMG (multi-channel)
 - Talk to Anweesha Sarkar about potentially building upon that project (jake)
 - Does she still have access to hardware
 - Can we meet with her in person
 - How did she implement the ML model into microprocessor
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 - Figure out how motor will interface with the ML and the EMG
 - See how the ML can be exported into the microprocessor
 - Brainstorm potential add-ons to existing hand
 - Optical mouse
 - Set up GitHub page for all data acquisition, signal analysis, machine learning, etc.
 - Provide a preliminary timeline for the project (Will be based off of Anweesha Sarkar conversation)
 - Apply to the IRB for waiver and permissions
 - Look for an industry product hand that we can buy/open source
 - Provide a semester timeline
 - Start assigning tasks to specific members and splitting into subgroups
 - Discuss PCB plans with possible mentors for PCB design help
 - Ensure that a custom PCB is realistic, reasonable, and worth the energy
- **Plans for the upcoming week**
- Logan – Figure out which sEMG to purchase and how to connect them to individual muscles. How many sensors do we need, start setting up simulation environment to test.
- Reid – Look into newer strategies for underactuation, determine if there is a link between how many motors we will use, and how many EMG sensors need to be bought. Provide a general timeline for the semester once direction is determined,
- Erik – Research EMG signal processing, test EMG sensors, familiarize myself with lab space, start 3D modeling, look into amputee hand/arm structure
- Adriana – Research correlation between hand movement and EMG sensors (# of sensors needed for each movement)

- Quinn – Discuss PCB concerns with potential mentors in the space. Figure out rough feasibility/cost estimations for PCB. Research how EMG signals can be processed and what hardware is necessary for machine learning capabilities + number of IO we're already planning to need.
- Redwan – Searching for microcontrollers to purchase, looking into EMG training data, setting up Git for our project
- Jacob – Look into motors and potential sensors that can be integrated into the hand. Research potential ways to actuate and reset finger movement. Find and submit a waiver for IRB so we can use human information. Look into open-source machine learning model for prosthetic hands.
- **Summary of weekly advisor meeting**
 - Went over scope of project that was discussed in Friday's meeting
 - There are open-sourced code models for different gestures from EMG (Github) (hand position prediction model, multiple EMG,
 - Reach out to Quinn's professor to aid in PCB design
 - Reach out to Professor Sarkar about using components of a previous group
 - Discussed possible implications of building upon a previous year's projects
 - Identified which EMG sensor could work with our project
 - Overall, we are reaching out to last year's group mentor to see if we can add onto their project, or if we continue with our current project scope
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